University of Toronto Mississauga
Department of Mathematical and Computational Sciences

Applied Time Series Analysis
STA457H5S, Winter 2022
Dr. Masoud Ataei

Assignment 1 Solutions

Due February 11, 2022 at 11:59 pm

Problem 1 (20 pts). Derive the mean, variance, autocovariance function and autocorrelation function for the following second order moving average model

$$MA(2): X_t = \mu + W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2},$$

where $\{W_t\} \overset{iid}{\sim} wn(0, \sigma_w^2)$. Note that you are NOT allowed to base your proof on the stationarity assumption.

Solution. Note that the mean of the process is

$$\mathbb{E}[X_t] = \mathbb{E}[\mu + W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2}] = \mu.$$

So, the autocovariance of X_t becomes

$$\begin{aligned}\gamma_{t,t-h} &= \mathbb{E}\left[(X_t - \mathbb{E}[X_t])\left(X_{t-h} - \mathbb{E}[X_{t-h}]\right)\right] \\ &= \mathbb{E}\left[(W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2})\left(W_{t-h} + \theta_1 W_{t-h-1} + \theta_2 W_{t-h-2}\right)\right]\end{aligned}$$

By noting that

$$\mathbb{E}[W_t W_s] = \begin{cases} \sigma_w^2 & \text{if } t = s \\ 0 & \text{if } t \neq s \end{cases}$$

and

$$\mathbb{E}[W_t^2] = Var[W_t] - (\mathbb{E}[W_t])^2 = \sigma_w^2 - 0 = \sigma_w^2,$$

we can compute $\gamma_{t,t-h}$ for different values of lag parameter h as follows:

(i) $h = 0:$

$$\begin{aligned}\gamma_{t,t} &= \mathbb{E}\left[(W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2})^2\right] \\ &= \mathbb{E}\left[W_t^2 + \theta_1^2 W_{t-1}^2 + \theta_2^2 W_{t-2}^2 + 2\theta_1 W_t W_{t-1} + 2\theta_1\theta_2 W_{t-1}W_{t-2} + 2\theta_2 W_t W_{t-2}\right] \\ &= (1+\theta_1^2+\theta_2^2)\mathbb{E}[W_t^2] = (1+\theta_1^2+\theta_2^2)\sigma_w^2\end{aligned}$$

(ii) $h = 1:$

$$\begin{aligned}\gamma_{t,t+1} &= \mathbb{E}\left[(W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2})(W_{t-1} + \theta_1 W_{t-2} + \theta_2 W_{t-3})\right] \\ &= \theta_1 \mathbb{E}[W_{t-1}^2] + \theta_1\theta_2 \mathbb{E}[W_{t-2}^2] = \theta_1(1+\theta_2)\sigma_w^2\end{aligned}$$

(iii) $h = 2:$

$$\begin{aligned}\gamma_{t,t+2} &= \mathbb{E}\left[(W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2})(W_{t-2} + \theta_1 W_{t-3} + \theta_2 W_{t-4})\right] \\ &= \theta_2 \mathbb{E}[W_{t-2}^2] = \theta_2 \sigma_w^2\end{aligned}$$

(iv) $h \geq 3:$

$$\gamma_{t,t-h} = 0.$$

The variance of the process would be as same as $\gamma_{t,t}$ and the autocorrelation function can be derived by dividing the value of autocovariance function at each lag by the value of the variance. That is,

$$\rho_{t,t-h} = \frac{\gamma_{t,t-h}}{\gamma_{t,t}}.$$

Hence,

$$\rho_{t,t-h} = \begin{cases} 1 & \text{if } h = 0 \\ \dfrac{\theta_1(1+\theta_2)}{1+\theta_1^2+\theta_2^2} & \text{if } h = 1 \\ \dfrac{\theta_2}{1+\theta_1^2+\theta_2^2} & \text{if } h = 2 \\ 0 & \text{if } h \geq 3 \end{cases}$$

Problem 2 (20 pts). Under the assumption of weak stationarity, derive the autocovariance function for the following second order autoregressive model

$$AR(2) : X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t,$$

where $\{W_t\} \overset{iid}{\sim} wn(0, \sigma_w^2)$.

Solution. The mean of the process is

$$\mathbb{E}[X_t] = \mathbb{E}[\phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t] = \phi_1 \mathbb{E}[X_{t-1}] + \phi_2 \mathbb{E}[X_{t-2}] + \mathbb{E}[W_t].$$

However, since the process $\{X_t\}$ is stationary, $X_t = X_{t-1} = X_{t-2} = \mu$. So, by replacing μ back in the equation above, we obtain

$$\mu = \phi_1 \mu + \phi_2 \mu \implies \mu = 0.$$

By noting that $\gamma_h = \gamma_{-h}$,

$$\mathbb{E}[W_t W_s] = \begin{cases} \sigma_w^2 & \text{if } t = s \\ 0 & \text{if } t \neq s \end{cases}$$

and

$$\mathbb{E}[W_t^2] = Var[W_t] - (\mathbb{E}[W_t])^2 = \sigma_w^2 - 0 = \sigma_w^2,$$

the autocovariance of X_t for $h \geq 1$ is obtained as follows:

$$\begin{aligned} \gamma_h &= \mathbb{E}\left[(X_t - \mathbb{E}[X_t])\,(X_{t-h} - \mathbb{E}[X_{t-h}])\right] \\ &= \mathbb{E}[X_t X_{t-h}] \\ &= \mathbb{E}\left[(\phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t) X_{t-h}\right] \\ &= \phi_1 \mathbb{E}[X_{t-1} X_{t-h}] + \phi_2 \mathbb{E}[X_{t-2} X_{t-h}] + \mathbb{E}[W_t X_{t-h}] \end{aligned}$$

Thus,

$$\gamma_h = \begin{cases} \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2} & \text{if } h \neq 0 \\ \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2} + \sigma_w^2 & \text{if } h = 0 \end{cases}$$

The initial conditions are then obtained by solving the following system of equations

$$\begin{cases} \gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma_w^2 \\ \gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1 \\ \gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0 \end{cases}$$

which yields

$$\gamma_0 = (\frac{1 - \phi_2}{1 + \phi_2}) \frac{\sigma_w^2}{(1 - \phi_2)^2 - \phi_1^2}$$

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2} \gamma_0 = (\frac{\phi_1}{1 + \phi_2}) \frac{\sigma_w^2}{(1 - \phi_2)^2 - \phi_1^2}$$

Lastly, given γ_0 and γ_1, we obtain γ_h as follows:

$$\gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}$$

Problem 3 (20 pts). Considering $\{W_t\} \overset{iid}{\sim} N(0, \sigma_w^2)$, for each of the following models derive the mean and autocovariance function, and discuss whether the model is stationary or not.

1. $X_t = W_t - W_{t-3}$
2. $X_t = W_3$
3. $X_t = t + W_3$
4. $X_t = W_t^2$
5. $X_t = W_t W_{t-2}$

Solution.

1- The process is stationary since

$$\mathbb{E}[X_t] = \mathbb{E}[W_t] - \mathbb{E}[W_{t-3}] = 0$$

which is a constant. Also,

$$\begin{aligned}\gamma_{t,s} &= \mathbb{E}[X_t X_s] = \mathbb{E}[W_t W_s] + \mathbb{E}[W_{t-3} W_s] + \mathbb{E}[W_t W_{s-3}] + \mathbb{E}[W_{t-3} W_{s-3}] \\ &= \mathbb{1}_{t=s} + \mathbb{1}_{t-3=s} + \mathbb{1}_{t=s-3} + \mathbb{1}_{t-3=s-3} \\ &= 2.\mathbb{1}_{|t-s|=0} + \mathbb{1}_{|t-s|=3}\end{aligned}$$

which is a function of $|t-s|$ (or h).

2- The process is stationary since

$$\mathbb{E}[X_t] = \mathbb{E}[W_3] = 0$$

and

$$\gamma_{t,s} = \mathbb{E}[X_t X_s] = \mathbb{E}[W_3^2] = \sigma_w^2$$

which are constants.

3- The process is not stationary since its mean is not constant, i.e., $\mathbb{E}[X_t] = t$.

4- First note that

$$W_t \sim N(0, \sigma_w^2) \implies \frac{W_t}{\sigma_w} \sim N(0,1) \implies \left(\frac{W_t}{\sigma_w}\right)^2 \sim \chi^2_{(1)}$$

From here we obtain

$$\frac{1}{\sigma_w^2}\mathbb{E}[W_t^2] = \mathbb{E}[\chi^2_{(1)}] = 1 \implies \mathbb{E}[W_t^2] = \sigma_w^2$$

Also,

$$\frac{1}{\sigma_w^4}Var[W_t^2] = Var[\chi^2_{(1)}] = 2 \implies Var[W_t^2] = 2\sigma_w^4$$

Then, the autocovariance function can be derived as follows:

$$\begin{aligned}
\gamma_{t,s} =& \mathbb{E}\left[(X_t - \sigma_w^2)(X_s - \sigma_w^2)\right] = \mathbb{E}[X_t X_s] - \sigma_w^2(\mathbb{E}[X_t] + \mathbb{E}[X_s]) + \sigma_w^4 \\
=& \mathbb{E}[W_t^2 W_s^2] - 2\sigma_w^4 + \sigma_w^4 \\
=& \mathbb{E}[W_t^2]\mathbb{E}[W_s^2] - \sigma_w^4 = \sigma_w^4 - \sigma_w^4 = 0
\end{aligned}$$

As it is seen, both the mean function and the autocovariance function are constants, hence the stochastic process is a stationary one.

The following formula to compute the moments of a random variable $W \sim \chi^2_{(\nu)}$ is one the most useful tools when dealing with Gaussian processes:

$$\mathbb{E}\left[W^\alpha\right] = \frac{\Gamma\left(\frac{\nu}{2} + \alpha\right)}{\Gamma\left(\frac{v}{2}\right)} 2^\alpha$$

such that the following condition is satisfied

$$\nu > -2\alpha.$$

5- The process is stationary since

$$\mathbb{E}[X_t] = \mathbb{E}[W_t]\mathbb{E}[W_{t-2}] = 0$$

which is a constant. Also,

$$\gamma_{t,s} = \mathbb{E}[X_t X_s] = \mathbb{E}[W_t W_{t-2} W_s W_{s-2}] = \sigma_w^4 \mathbb{1}_{t=s}$$

which is the function of the lag parameter $h = s - t$. So, the underling stochastic process is stationary.

Problem 4 (20 pts). Suppose that U_t and V_t are independent and identically distributed (iid) sequences, with $\mathbb{P}[U_t = 0] = \mathbb{P}[U_t = 1] = \frac{1}{2}$ and $\mathbb{P}[V_t = -1] = \mathbb{P}[V_t = 1] = \frac{1}{2}$. Define the time series model

$$X_t = U_t(1 - U_{t-1})V_t.$$

Show that $\{X_t\}$ is a white noise process, whereas it is not necessarily iid, i.e., $\{X_t\} \sim wn(0, \sigma_w^2)$ but not $\{X_t\} \overset{iid}{\sim} wn(0, \sigma_w^2)$.

Solution. Suppose $\{U_t\}$ and $\{V_t\}$ are independent and identically distributed (iid) sequences, with the probability mass functions given above. To check whether $\{X_t\}$ is white noise, we need to compute its mean and covariances. For the mean

$$\mathbb{E}[X_t] = \mathbb{E}[U_t(1 - U_{t-1})]V_t = \mathbb{E}[U_t](1 - \mathbb{E}[U_{t-1}])(\mathbb{E}[V_t]) = 0$$

For the autocovariances, we have

$$\gamma_{t,s} = \mathbb{E}\left[(U_t(1 - U_{t-1}))W_t U_s(1 - U_{s-1})W_s\right] = \mathbb{E}[U_t(1 - U_{t-1})U_s(1 - U_{s-1})]\mathbb{E}[V_t V_s]$$

If $t \neq s$ then the last term $\mathbb{E}[V_t V_s] = \mathbb{E}[V_t]\mathbb{E}[V_s] = 0$. Therefore, $\{X_t\}$ is uncorrelated.

If $t = s$ then $\mathbb{E}[V_t V_s] = \mathbb{E}[V_t^2] = 1$, so

$$\gamma_{t,t} = \mathbb{E}[\mathbb{U}_t^2(1 - U_{t-1})^2] = \frac{1}{4}$$

which implies that $\{X_t\}$ has constant variance. Hence it is a white noise process.
To show that $\{X_t\}$ is not iid, note that $X_{t-1} = 1$ implies $U_{t-1} = 1$, which in turn leads to $X_t = 0$. Therefore,

$$\mathbb{P}[X_{t-1} = 1, X_t = 1] = 0.$$

Since this is not equal to $\mathbb{P}[X_{t-1} = 1]\mathbb{P}[X_t = 1] = \frac{1}{64}$, the X_t and X_{t-1} are not independent.

Problem 5 (20 pts). Each column of the "`Dataset.xlsx`" corresponds to a realization of some stochastic process. Sketch the time plot and correlogram for each one of the time series using the command

$$\texttt{acf}(x, \text{lag.max} = 20).$$

Interpret the correlogram of each realized time series data and then decide which one of the $MA(1)$, $MA(2)$, $AR(1)$, RW (with or without drift), or white noise processes are most suitable to model the underlying stochastic processes. You should present your final models by writing their corresponding equations explicitly.

Find a relatively reasonable set of estimates for the parameters of each model by *trial and error*, e.g., for a candidate AR(1) model you may plug in different values for ϕ_1 and σ_w^2, and thereafter compare the plot of the computed sample ACF of your time series against the plot of the theoretical ACF of your proposed $AR(1)$ process.

Please note that there is no need to submit your R implementations. However, it is important to demonstrate the procedure by which you have carried out the trial and error step and eventually obtained the estimates for parameters of each model. It will be further informative to present a figure comprised of three horizontal subplots: 1- time plot, 2- correlogram, and 3- theoretical ACF (sketched according to the estimates of parameters obtained earlier) for each process.

Solution. The data generating processes (true underlying models) are:

$$x1 : X_t = W_t$$

$$x2 : X_t = 0.8X_{t-1} + W_t$$

$$x3 : X_t = -0.8X_{t-1} + W_t$$

$$x4 : X_t = W_t - 0.8W_{t-1}$$

$$x5 : X_t = W_t + 0.7W_{t-1} - 0.2W_{t-2},$$

where $W_t \overset{iid}{\sim} N(0, 1)$.

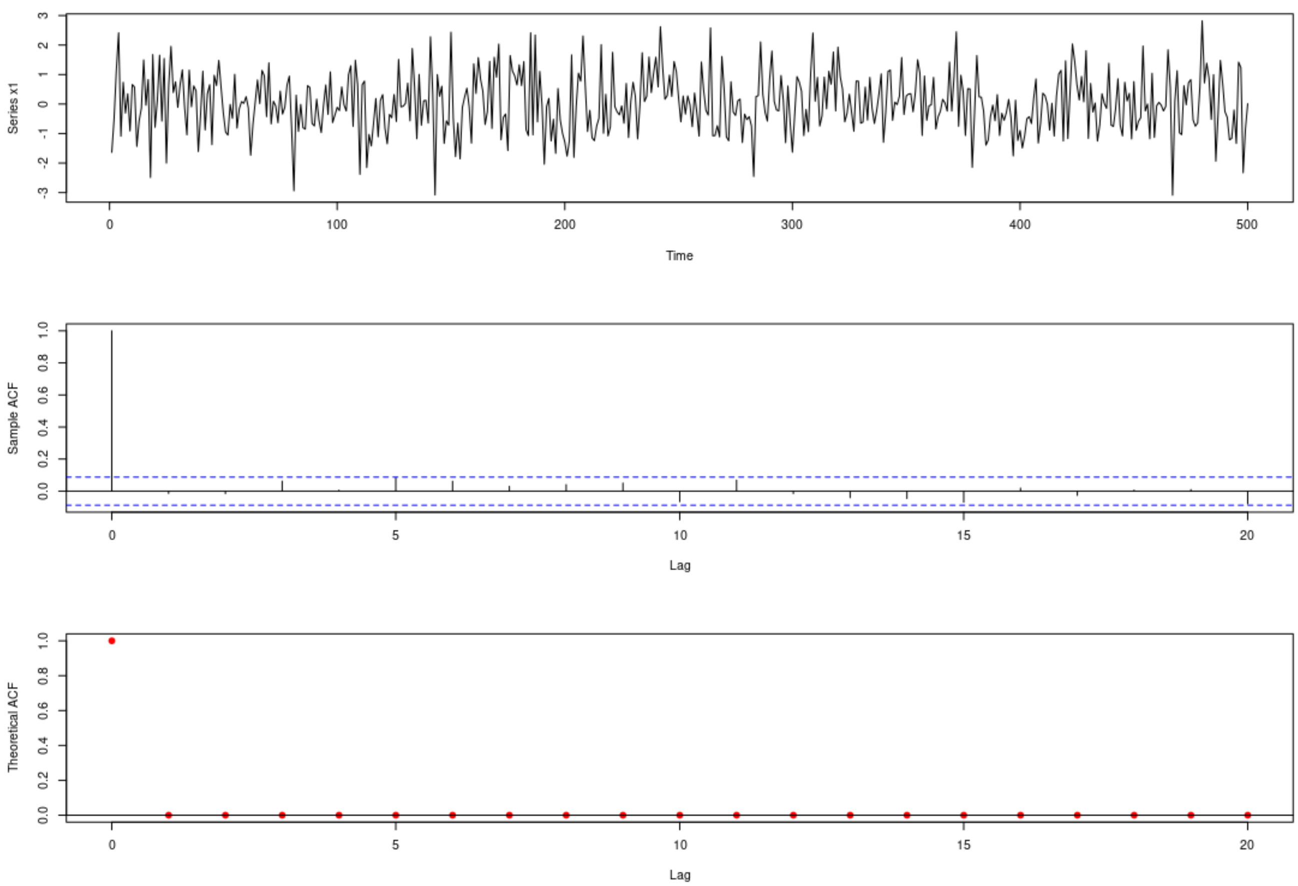

Figure 1: Figures for time series x1.

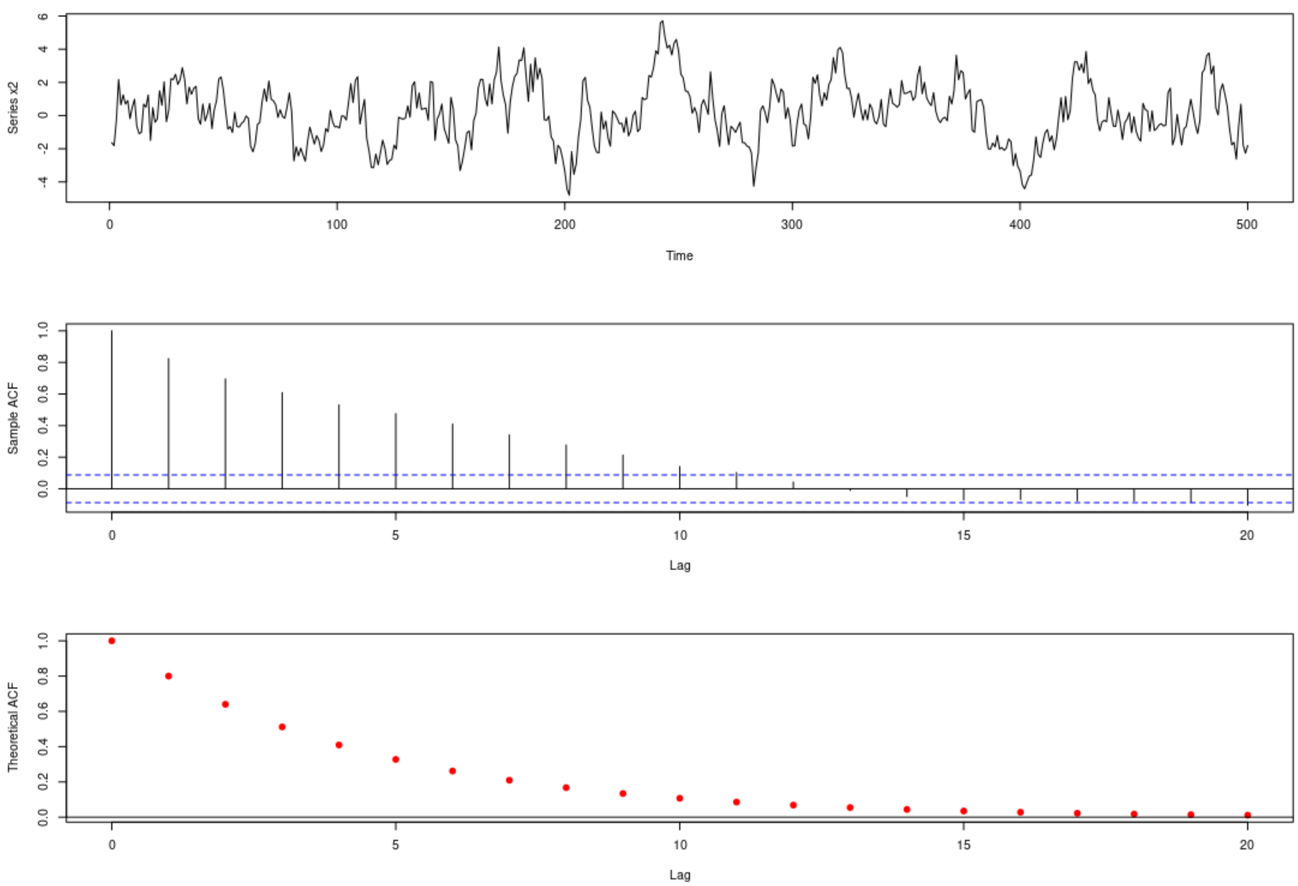

Figure 2: Figures for time series x2.

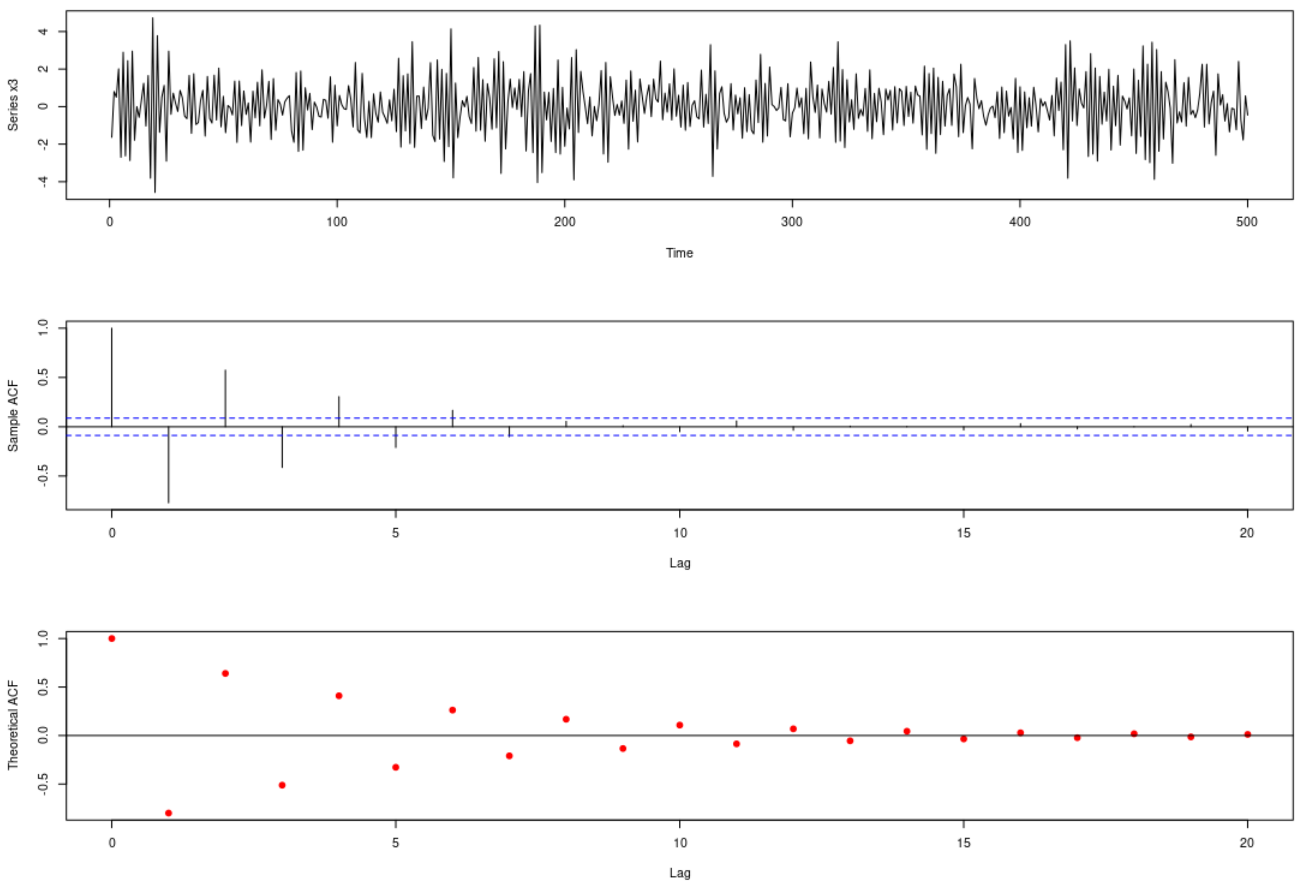

Figure 3: Figures for time series x3.

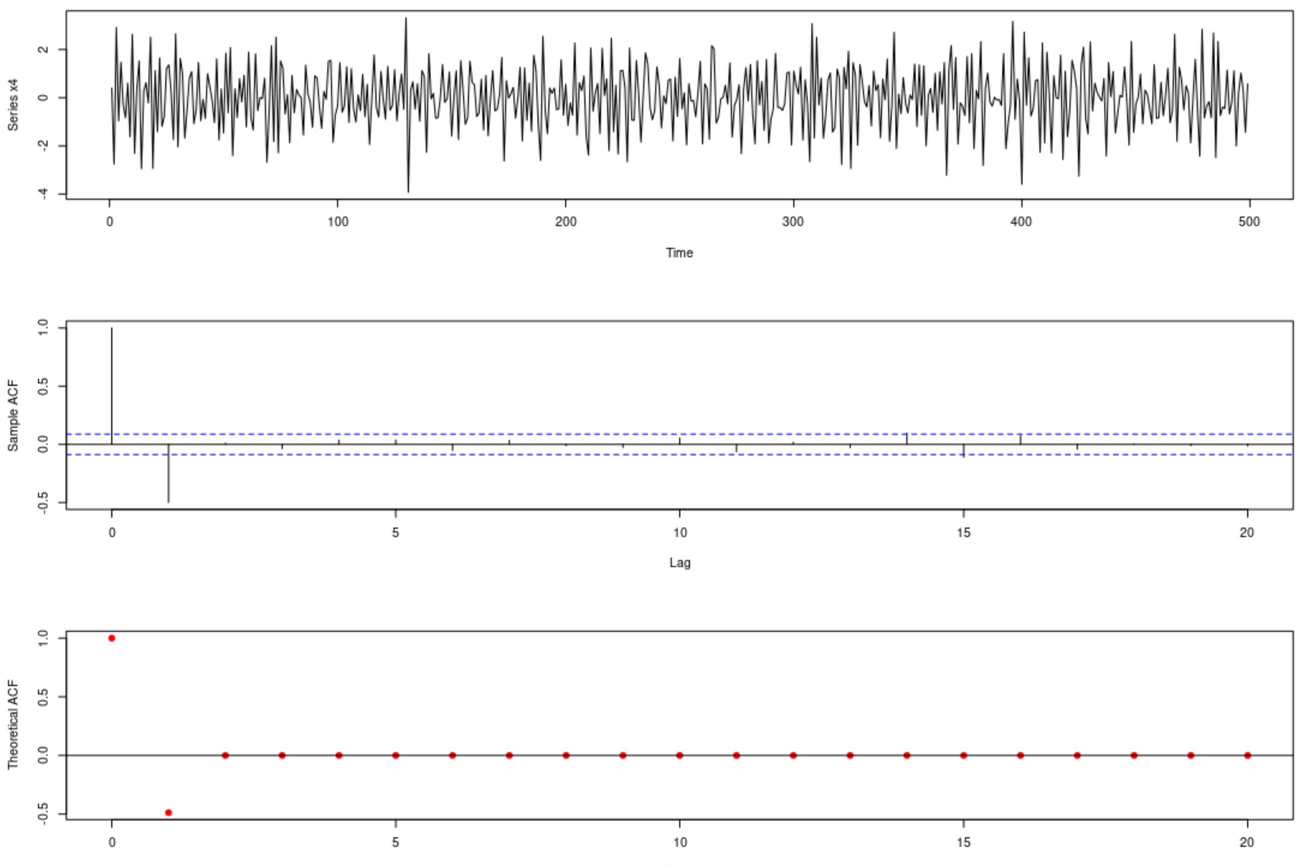

Figure 4: Figures for time series x4.

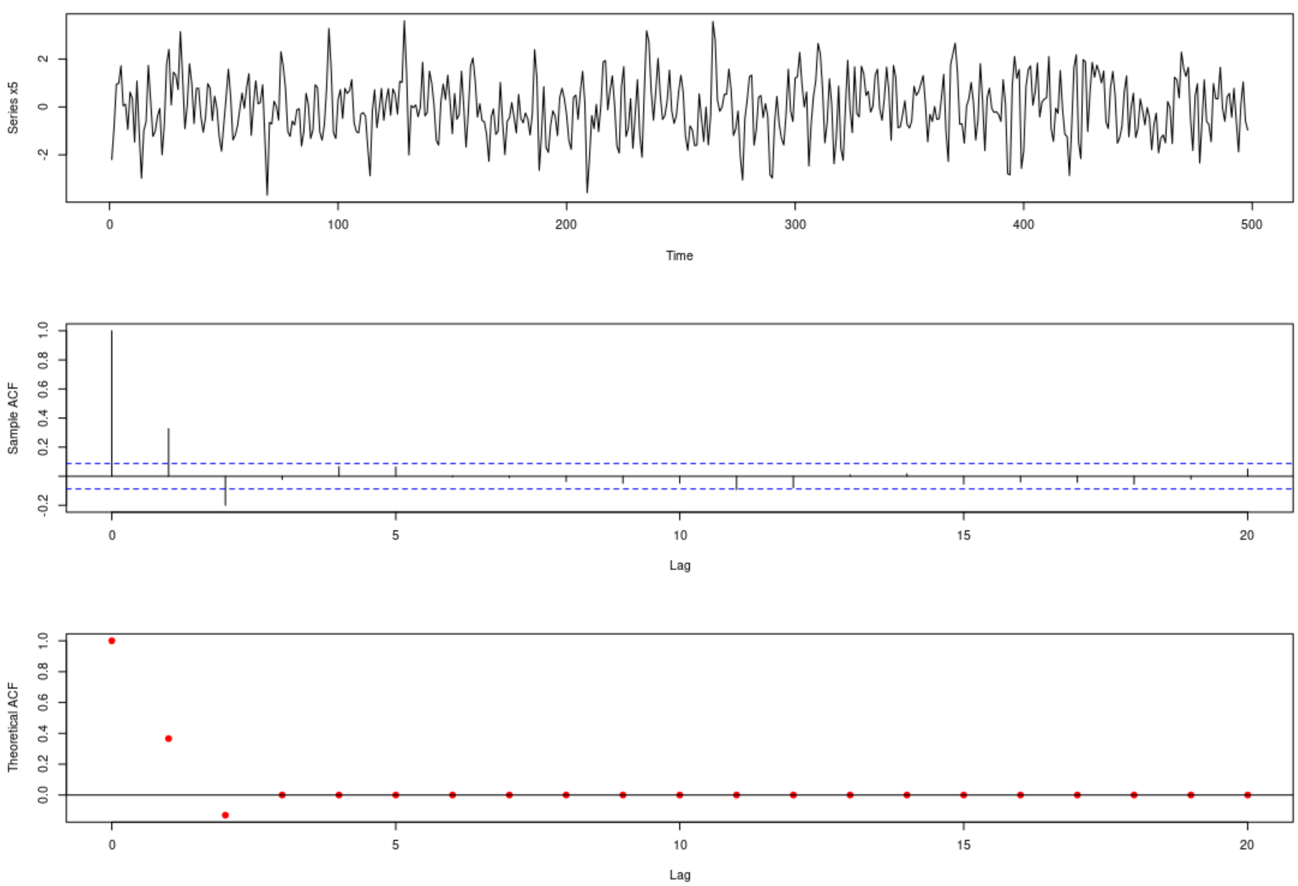

Figure 5: Figures for time series x5.

```r

library("readxl")

df <- read_excel("/home/masoud/Documents/UofT/Resources/My_Course/Assignments/Assignment_1/Data.xlsx")

####################################################################################################

n = 20
x <- 0:n
y <- rep(0,n+1)
y[1] = 1

par(mfrow=c(3,1),mar=c(4,4,4,4))
plot(df$x1, type="l", xlab="Time", ylab="Series x1")
acf(df$x1,20,xlab="Lag",ylab="Sample ACF",main="")
plot(x,y, pch = 19,col="red",xlab="Lag",ylab="Theoretical ACF",main="")
abline(h=0, col="black")

####################################################################################################

n = 20
x <- 0:n
y <- 0.8^x

par(mfrow=c(3,1),mar=c(4,4,4,4))
plot(df$x2, type="l", xlab="Time", ylab="Series x2")
acf(df$x2,20,xlab="Lag",ylab="Sample ACF",main="")
plot(x,y, pch = 19,col="red",xlab="Lag",ylab="Theoretical ACF",main="")
abline(h=0, col="black")

####################################################################################################

n = 20
x <- 0:n
y <- (-0.8)^x

par(mfrow=c(3,1),mar=c(4,4,4,4))
plot(df$x3, type="l", xlab="Time", ylab="Series x3")
acf(df$x3,20,xlab="Lag",ylab="Sample ACF",main="")
plot(x,y, pch = 19,col="red",xlab="Lag",ylab="Theoretical ACF",main="")
abline(h=0, col="black")

####################################################################################################

n = 20
x <- 0:n
y <- rep(0,n+1)
theta <- -0.8
y[1] = 1
y[2] = theta/(1+theta^2)

par(mfrow=c(3,1),mar=c(4,4,4,4))
plot(df$x4[1:499], type="l", xlab="Time", ylab="Series x4")
acf(df$x4[1:499],20,xlab="Lag",ylab="Sample ACF",main="")
plot(x,y, pch = 19,col="red",xlab="Lag",ylab="Theoretical ACF",main="")
abline(h=0, col="black")

####################################################################################################

n = 20
x <- 0:n
y <- rep(0,n+1)
theta1 <- 0.7
theta2 <- -0.2
y[1] = 1
y[2] = theta1*(1+theta2)/(1+theta1^2+theta2^2)
y[3] = theta2/(1+theta1^2+theta2^2)

par(mfrow=c(3,1),mar=c(4,4,4,4))
plot(df$x5[1:498], type="l", xlab="Time", ylab="Series x5")
acf(df$x5[1:498],20,xlab="Lag",ylab="Sample ACF",main="")
plot(x,y, pch = 19,col="red",xlab="Lag",ylab="Theoretical ACF",main="")
abline(h=0, col="black")
```

Figure 6: The accompanied R codes.